

Jann Paul Mattern

Curriculum Vitae

Ocean Sciences Department
University of California Santa Cruz
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Applied Statistician specializing in creating and employing advanced statistical methodologies to analyze large datasets and complex numerical models. Expertise encompasses areas such as data assimilation, optimization, and uncertainty analyses. My current research is dedicated to understanding ocean biogeochemical cycles, including carbon dynamics, and examining the local transmission patterns of Covid-19.

research

PhD thesis *Parameter, State and Uncertainty Estimation for 3-dimensional Biological Ocean Models* (advisors: Dr. Katja Fennel and Dr. Mike Dowd)

- data assimilation for complex, high-dimensional models and satellite observations using particle filter and statistical emulator techniques
- assessment of model uncertainty and sensitivity
- application of the methods to regional ROMS models for the Middle-Atlantic Bight and the Texas-Louisiana Shelf

MSc thesis *Ensemble-based data assimilation for a physical-biological ocean model near Bermuda* (advisors: Dr. Dirk Langemann, *Universität zu Lübeck*, Dr. Katja Fennel and Dr. Mike Dowd, *Dalhousie University*)

- implementation and comparison of the Ensemble Kalman Filter and a particle filter for a 1-dimensional ocean model and real-world observations

BSc thesis *Unterschriftsverifikation basierend auf lokal extrahierten Merkmalen temporaler Sequenzen* (*Verification of handwritten signatures based on locally extracted features of temporal sequences*; advisors: Dr. Thomas Martinetz and Dipl. Inf. Kai Labusch)

- development of a method to compare and classify handwritten signatures using sequence alignment and optimization techniques

academic work experience

since Sep 2017 Research Scientist, Ocean Sciences Department, *UC Santa Cruz*

2013 – 2017 Postdoctoral Fellow, Ocean Sciences Department, *UC Santa Cruz*
(advisor: Dr. Christopher A. Edwards)

2012 – 2013 Postdoctoral Fellow, Department of Oceanography, *Dalhousie University*
(advisor: Dr. Katja Fennel)

2008 – 2008 Research assistant, Department of Oceanography, *Dalhousie University*

2004 – 2007 Student assistant, Institute for Neuro- and Bioinformatics, *Universität zu Lübeck*

education

- 2009 – 2012 PhD program: Statistics (with emphasis on Oceanography)
Dalhousie University, Halifax, Canada
- 2007 – 2008 Visiting student, Department of Oceanography as part of the MSc program
Dalhousie University, Halifax, Canada
- 2005 – 2008 MSc program: Computational Life Sciences, *Universität zu Lübeck*, Lübeck, Germany
- 2002 – 2005 BSc program: Computational Life Sciences, *Universität zu Lübeck*, Lübeck, Germany

selected peer-reviewed publications (full list on [Google Scholar](#))

- J. P. Mattern**, S. Dutkiewicz, J. E. Moscoso and C. A. Edwards (submitted to *Limnology & Oceanography*), Bottom-Up and Top-Down Controls on Diatom Emergence in the California Current System: Insights from a Size-Structured Regional Ecosystem Model.
- J. P. Mattern**, Y. Takeshita, C. Rocha and C. A. Edwards (submitted to *Biogeosciences*), Improving coastal ocean pH estimates through assimilation of glider observations and hybrid statistical methods.
- A. M. Kilpatrick, **J. P. Mattern**, B. J. Gardner, B. Shoemaker, S. Williams, D. Chavez, C. Gordon and D. Ghilarducci (in prep.), Is wastewater a leading indicator for respiratory syncytial virus, influenza A virus, or SARS-CoV-2?
- S. Dutkiewicz, C. L. Follett, M. J. Follows, F. Henderikx-Freitas, F. Ribalet, M. R. Gradoville, S. N. Coesel, H. Farnelid, Z. V. Finkel, A. J. Irwin, O. Jahn, D. M. Karl, **J. P. Mattern**, A. E. White, J. Zehr and E. V. Armbrust (2024), Multiple Biotic Interactions Set Phytoplankton Community Structure Across Environmental Gradients, *Limnology and Oceanography*. doi: [10.1002/lno.12555](#)
- K. Fennel, M. C. Long, C. Algar, B. Carter, D. Keller, A. Laurent, **J. P. Mattern**, R. Musgrave, A. Oschlies, J. Ostiguy, J. B. Palter and D. B. Whitt (2023), Modeling considerations for research on Ocean Alkalinity Enhancement (OAE), *State of the Planet*. doi: [10.5194/sp-2-oea2023-9-2023](#).
- J. P. Mattern** and C. A. Edwards (2023), Ensemble optimal interpolation for adjoint-free biogeochemical data assimilation, *PLOS ONE*. doi: [10.1371/journal.pone.0291039](#)
- J. G. Izett, **J. P. Mattern**, A. M. Moore and C. A. Edwards (2023), Evaluating Alternate Methods of 4D-Var Data Assimilation in a coupled hydrodynamic—Four-component biogeochemical model of the California Current System, *Ocean Modelling*. doi: [10.1016/j.ocemod.2023.102253](#)
- K. Fennel, **J. P. Mattern**, S. Doney, L. Bopp, A. M. Moore, B. Wang and L. Yu (2022), Ocean Biogeochemical Modelling, *Nature Reviews Methods Primers*. doi: [10.1038/s43586-022-00154-2](#)
- J. P. Mattern**, K. Glauning, G. L. Britten, J. Casey, S. Hyun, E. V. Armbrust, M. Follows, Z. Harchaoui and F. Ribalet (2022), A Bayesian approach to modeling phytoplankton population dynamics from size distribution time series, *PLOS Computational Biology*. doi: [10.1371/journal.pcbi.1009733](#)
- J. P. Mattern**, C. A. Edwards and C. N. Hill (2019), Dual number-based variational data assimilation – Constructing exact tangent linear and adjoint code from nonlinear model evaluations, *PLOS ONE*. doi: [10.1371/journal.pone.0223131](#)
- J. P. Mattern** and C. A. Edwards (2019), A simple difference quotient-based approximation for biogeochemical tangent linear and adjoint models, *Journal of Geophysical Research: Oceans*. doi: [10.1029/2018JC014283](#)
- M. Irie, F. Hirose, T. Okada, **J. P. Mattern** and K. Fennel (2018), Modeling of nitrogen and phosphorus profiles in sediment of Osaka Bay, Japan with parameter optimization using the polynomial chaos expansion, *Coastal Engineering Journal*. doi: [10.1080/21664250.2018.1531814](#).
- J. P. Mattern**, C. A. Edwards and A. M. Moore (2018), Improving variational data assimilation through background and observation error adjustments, *Monthly Weather Review*. doi: [10.1175/MWR-D-17-0263.1](#)

- J. P. Mattern**, H. Song, C. A. Edwards, A. M. Moore and J. Fiechter (2017), Data assimilation of physical and chlorophyll observations in the California Current System using two biogeochemical models, *Ocean Modelling*. doi: [10.1016/j.ocemod.2016.12.002](https://doi.org/10.1016/j.ocemod.2016.12.002)
- J. P. Mattern** and C. A. Edwards (2017), Simple parameter estimation for complex models – testing evolutionary techniques on 3-dimensional biogeochemical ocean models, *Journal of Marine Systems*. doi: [10.1016/j.jmarsys.2016.10.012](https://doi.org/10.1016/j.jmarsys.2016.10.012).
- A. Kuhn, K. Fennel and **J. P. Mattern** (2015), Model investigations of the North Atlantic spring bloom initiation, *Progress in Oceanography*. doi: [10.1016/j.pocean.2015.07.004](https://doi.org/10.1016/j.pocean.2015.07.004)
- J. P. Mattern**, K. Fennel and M. Dowd (2014), Periodic time-dependent parameters improving forecasting abilities of biological ocean models, *Geophysical Research Letters*. doi: [10.1002/2014GL061178](https://doi.org/10.1002/2014GL061178)
- J. P. Mattern**, M. Dowd, and K. Fennel (2013), Particle Filter-based Data Assimilation for a 3-dimensional Biological Ocean Model and Satellite Observations, *Journal of Geophysical Research*. doi: [10.1002/jgrc.20213](https://doi.org/10.1002/jgrc.20213)
- J. P. Mattern**, K. Fennel, and M. Dowd (2013), Sensitivity and Uncertainty Analysis of Model Hypoxia Estimates for the Texas-Louisiana Shelf, *Journal of Geophysical Research*. doi: [10.1002/jgrc.20130](https://doi.org/10.1002/jgrc.20130)
- J. P. Mattern**, K. Fennel, and M. Dowd (2012), Estimating time-dependent parameters for a biological ocean model using an emulator approach, *Journal of Marine Systems*. doi: [10.1016/j.jmarsys.2012.01.015](https://doi.org/10.1016/j.jmarsys.2012.01.015)
- A. J. Hidy, J. C. Gosse, J. L. Pederson, **J. P. Mattern**, and R. C. Finkel (2010), A geologically constrained Monte Carlo approach to modeling exposure ages from profiles of cosmogenic nuclides: An example from Lees Ferry, Arizona, *Geochemistry Geophysics Geosystems*. doi: [10.1029/2010GC003084](https://doi.org/10.1029/2010GC003084)

selected published abstracts

- J. P. Mattern**, C. A. Edwards, J. E. Moscoso, S. Dutkiewicz and O. Jahn (2024), Insights from ROMS-Darwin, a Regional Size-Spectral Biogeochemical Model. *2024 Ocean Sciences Meeting*, New Orleans, LA, USA. (poster presentation)
- J. P. Mattern**, C. A. Edwards, Y. Takeshita and J. Long (2023), Integrating autonomous pH observations and statistical estimates with dynamical ocean models *2023 Eastern Pacific Ocean Conference*, Fallen Leaf Lake, CA, USA. (oral presentation)
- J. P. Mattern** and C. A. Edwards (2022), A four-dimensional ensemble optimal interpolation approach for adjoint-free data assimilation in a regional biogeochemical ocean model. *2022 Joint EuroSea/OceanPredict Workshop*, Met Office, Exeter, UK. (oral presentation)
- J. P. Mattern** and C. A. Edwards (2022), A four-dimensional ensemble optimal interpolation approach for adjoint-free biogeochemical data assimilation. *2022 Ocean Sciences Meeting*, virtual. (oral presentation)
- J. P. Mattern**, C. A. Edwards and C. N. Hill (2020), Using dual numbers for automatic differentiation of complex functions – a simple way to create data assimilation code for coupled models. *2020 Ocean Sciences Meeting*, San Diego, CA, USA. (oral presentation)
- J. P. Mattern**, C. A. Edwards and C. N. Hill (2019), Two simple ways for creating model-independent biogeochemical tangent linear and adjoint code. *Ocean Predict '19*, Halifax, Canada. (oral presentation)
- J. P. Mattern**, C. A. Edwards and A. M. Moore (2018), Error covariance tuning for data assimilation in biogeochemical ocean models. *2018 Ocean Sciences Meeting*, Portland, OR, USA. (oral presentation)
- J. P. Mattern**, K. Fennel, M. Dowd, C. A. Edwards, H. Song, A. M. Moore and J. Fiechter (2015), When Ocean Models meet Observations – Data Assimilation in Marine Ecosystem Models. *Scripps Institution of Oceanography*, La Jolla, CA, USA. (invited seminar talk)

other professional activities

Modeling the outbreak of Covid-19 in the Santa Cruz County as a Volunteer with the County of Santa Cruz, California (since April 2020) and as a member of the University of California - CDPH (California Department of Public Health) Modeling Consortium for COVID-19 rapid analytic decision-making support (since October 2021). Principal developer of the Covid-19 model for Santa Cruz County, providing weekly model updates. Results are featured on the County of Santa Cruz Health Services Agency dashboard.

Member of the Computational Biogeochemical Modeling of Marine Ecosystems CBIOMES group.

Member of the Marine Ecosystem Analysis and Prediction Task Team MEAP-TT of the OceanPredict Initiative.

Co-organizer of the "Workshop on Bayesian Plankton Modeling" held at the Massachusetts Institute of Technology in January 2020.

Co-organizer of the "CBIOMES Workshop on Transects and Eco-provinces" held at the Massachusetts Institute of Technology in January 2023.

teaching experience	Teaching graduate level course "Python programming for Ocean Scientists" consisting of 19 classes UC Santa Cruz (2017; 28.5 h total) Teaching IAP class "Bayesian modeling of ecological systems using the "Stan" software package", Massachusetts Institute of Technology (2020; 4h total) Teaching labs for "Marine Modelling", Dalhousie University (2008, 2010, 2011; 9h total) Teaching class in "Multivariate Analysis", Dalhousie University (2011; 1.5h total) Successfully completed the 2016 Professional Development Program, that prepares scientists and engineers to be leaders in effective and inclusive education.
manuscript reviewer	Biogeosciences (18 reviews), Frontiers in Earth Sciences (1 review), Frontiers in Marine Sciences (4 reviews), Geoscientific Model Development (6 reviews), Great Lakes Research (2 reviews), Hydrology and Earth System Sciences (2 reviews), Journal of Agricultural, Biological, and Environmental Statistics (1 review), Journal of Advances in Modeling Earth Systems (2 reviews), Journal of Geophysical Research (7 reviews), Journal of Marine Systems (5 reviews), Monthly Weather Review (6 reviews), Nature Ecology & Evolution (2 reviews), Ocean Modelling (15 reviews), Ocean Science Discussions (6 reviews), Progress in Oceanography (1 review), Quarterly Journal of the Royal Meteorological Society (2 reviews), Remote Sensing Letters (2 reviews), State of the Planet (1 review), Water (2 reviews)

computer/programming skills

advanced experience:	Python, bash, Stan, Matlab	intermediate experience:	Julia, FORTRAN, C++, Java, JavaScript
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